Homework 5

Due March 3 at 11 pm

Unless stated otherwise, justify any answers you give. You can work in groups, but each student must write their own solution based on their own understanding of the problem. Please submit each problem on a separate page (i.e., 1a and 1b can be on the same page but 1 and 2 must be on different pages).

1. (Modified bootstrap Gaussian confidence interval)

   (a) Explain how to modify the bootstrap Gaussian confidence interval for the correlation coefficient using Fisher's transformation.

   (b) Apply your method on the correlation between *footlength* and *stature* in `ANSUR II MALE Public.csv` to compute 0.95 confidence intervals from n samples. Plot them along with non-modified 0.95 bootstrap Gaussian confidence intervals and also with the population correlation coefficient. You can try different sample sizes $n \in \{25, 100\}$.

   (c) Estimate and report the probability that the population correlation belongs to each type of confidence interval.

2. (Confidence intervals for the variance)

   (a) Explain how to build a $1 - \alpha$ bootstrap Gaussian confidence intervals for the population variance from n data points.

   (b) Explain how to build a $1 - \alpha$ bootstrap percentile confidence intervals for the population variance from n data points.

   (c) Build 0.95 bootstrap Gaussian and percentile confidence intervals for the population variance of *stature* in `ANSUR II MALE Public.csv` setting $n = 100$. Plot the confidence intervals along with the true population variance.

   (d) Estimate and report the probability that the population variance belongs to each type of confidence interval.